

Data, Not Architecture: An Empirical Study of Replay-Based Continual Learning for Video Action Recognition

coad-mini · Something-Something V2 (48 classes, 8 sequential stages) · CLIP + GRU + A-GEM
[Name] · [Student ID] · [Course] · 2026

Problem

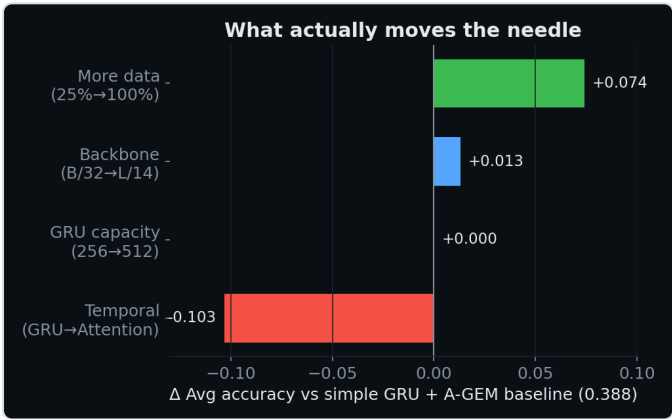
When a video action recognizer learns new action classes *sequentially* (in stages), it **catastrophically forgets** earlier ones. We study how to mitigate this forgetting, and — once mitigated — what the *next* bottleneck to accuracy actually is.

Approach

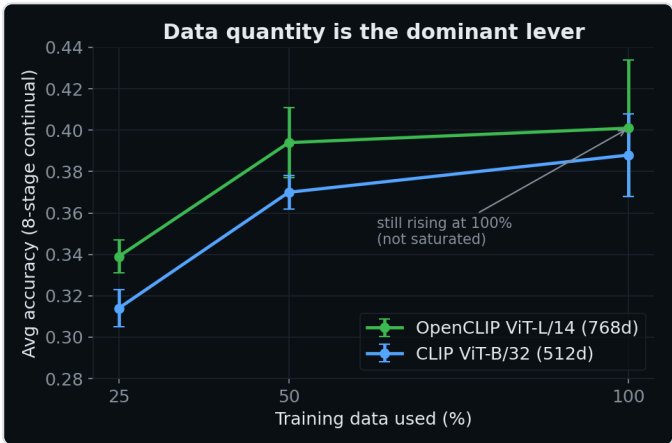
A compact pipeline trained stage by stage; **A-GEM** projects each gradient so updates do not increase loss on a replay memory of past classes.

video → 16 uniform frames → CLIP ViT-B/32 (512-d) → GRU classifier → 8-stage continual evaluation

Key results



Effect of each lever on Avg Acc (vs simple GRU + A-GEM).



Accuracy vs training-data fraction (B/32 vs L/14).

- A-GEM beats sequential training:** +14.1%p Avg Acc and +15.7%p Stage-1 forgetting protection (48 classes / 8 stages, 3 seeds); the gain grows with more classes/stages.
- Source of the gain:** class-balanced memory selection + a 25% replay ratio.
- Systematic bottleneck study** (all controlled, same setup):

Lever	Δ Avg Acc	Verdict
GRU capacity (256→512, +layers)	~0.000	no gain
Backbone (CLIP B/32 → OpenCLIP L/14)	+0.013	within noise
Temporal model (GRU → GRU + Attention)	-0.103	worse (+forgetting)
Training data (25% → 100%)	+0.06–0.07	dominant lever

Conclusion. A simple **GRU + A-GEM** is the robust optimum for this setting; further accuracy comes from **data quantity, not architectural complexity**. The data-scaling curves are still rising at 100% — i.e. not yet saturated.

System / demo

On top of the model we built an interactive demo (live): realtime captioning (Baseline vs A-GEM, upload or webcam), **few-shot enrollment** — teach a brand-new action from a few clips that appears in the live caption instantly without forgetting the rest — and OOD / anomaly detection.